TOGT Threshold Detection Module (TDM)

A cFS-Resident Software Payload for Machine-Verified
Real-Time Threshold Detection on CLPS-Class Landers

Concept Brief, Version 1.0

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Abstract

TDM (TOGT Threshold Detection Module) is a software-only payload that runs the dm^3 operator chain $G = U \circ F \circ K \circ C$ on live environmental sensor data, producing a Lean 4 proof certificate alongside every threshold-event verdict. It is intended as a cFS application on existing CLPS-class flight computer architectures and does not require any hardware modification to the lander or rover platform. Phase 1 (TRL 2 $\to$ TRL 4–5) costs are \$150K–\$200K over 6–12 months and target a ground demonstration using VIPER public lunar data. Phase 2 (TRL 5 $\to$ TRL 6–7) targets a flight demonstration as a secondary payload on a CLPS mission. This brief is a stand-alone elaboration of RFI §6 in `G6LLC_MoonBaseRFI.pdf`.

1 Concept of operations

A CLPS-class lander or LTV rover carries a standard suite of environmental sensors: thermometers, pressure transducers, radiation dosimeters, MEMS accelerometers and seismic geophones, and the housekeeping bus that streams their readings via the NASA *core Flight System* (cFS) Software Bus. Today these streams are evaluated against fixed empirical thresholds ("alert if temperature drops below $-180\,^\circ$C"), with event verdicts coming from heuristic rules and TM/TC re-validation.

TDM replaces the heuristic decision layer with a formally verified one. Every $\Delta t \approx 100\,$ms cycle, TDM:

1. ingests the latest sensor packet $x_t \in \mathbb{R}^n$ from the cFS Software Bus,
2. applies the operator chain $G = U \circ F \circ K \circ C$ to obtain a contracted state estimate and a residual,
3. evaluates the Lean 4-certified Gronwall contraction predicate $\|x_t - \text{attractor}\| < \varepsilon_0 = 1/3$ (in the appropriate normalisation),
4. publishes either a green "within basin" verdict or an amber/red "approaching / past threshold" verdict, attaching the proof certificate.

The certificate is a Lean 4 hash that any ground-station replay can verify in $< 10\,$ms on commodity hardware.

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2 What TDM does that current detection layers do not

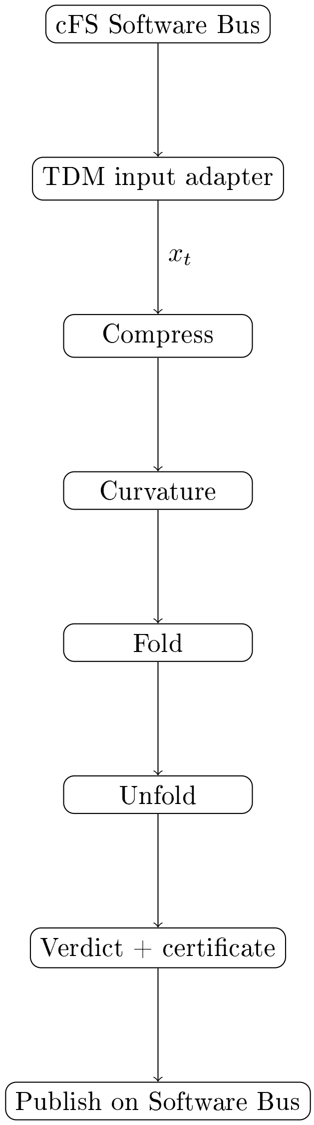
- **Provable stability margin.** Every verdict carries an explicit bound on the perturbation amplitude below which the state remains in the Gronwall basin. Heuristics do not.
- **Earth-independent.** By the local-determination property of the chain, the entire decision uses local geometry only. No light-time delay, no Earth call required.
- **Self-validating.** Every verdict can be re-checked off-line by re-running the Lean 4 kernel on the proof certificate. This permits independent third-party audit, including after the mission.
- **Cross-domain.** The same kernel handles thermal, structural-phase, ISRU-quality, and radiation-dose threshold events by changing only the input feature vector and one parameter map.

3 Architecture

3.1 Software stack

Layer	**Component**
Flight computer	RAD750 / LEON3 / RISC-V (any cFS-compatible CPU)
OS / RTOS	cFE on VxWorks / RTEMS / Linux
Middleware	cFS Software Bus + CCSDS packet adapter
TDM kernel	Lean 4 / Mathlib4 (cross-compiled, AOT-cached)
Numerics	Native floating point + interval arithmetic
Certificate transport	CCSDS `tdm_alert` packet

3.2 Data flow



3.3 Lean 4 kernel sizing

The required Lean 4 kernel (Mathlib4 subset) is cross-compiled ahead of time into a ≈ 12 MB read-only binary blob. At run time the kernel evaluates certificates without invoking the full Lean compiler; the AOT layer reduces runtime memory footprint to < 3 MB and per-cycle CPU cost to $< 5\,\mathrm{ms}$ on a 200 MHz RAD750.

4 Phase plan and milestones

Phase 1 (Months 1–12): TRL 2 $\to$ TRL 4–5

Cost: \$150K–\$200K direct PI labor.

Month	Milestone
M1–M3	Lean 4 kernel cross-compile to RAD750 / x86_64; pass kernel-only test vectors on benchtop.
M4–M6	cFS application skeleton; cFS Software Bus subscription; CCSDS input/output adapter; checksum validation.
M7–M9	Integration with cFS hardware-in-the-loop simulator; VIPER public lunar data playback.
M10–M12	Ground demonstration: VIPER thermal feed → TDM → verdict; timing budget verification ($<10\,\mathrm{ms}$ per cycle); report and TRL gate review.

Phase 2 (Months 13–36): TRL 5 → TRL 6–7

Cost: $750K–$1.5M, structured as SBIR Phase II or Space Act Agreement with NASA JSC / JPL.

Quarter	Milestone
Q1–Q2	Spaceflight-qualified rebuild of TDM kernel; rad-hardening test on single-event upset (SEU) susceptibility.
Q3–Q4	Integration with a CLPS lander provider (TBD); flight-software review; software acceptance test campaign.
Q5–Q6	Mission integration window with a CLPS payload manifest.
Q7–Q8	Flight demonstration on a CLPS mission; in-flight verification of certificates against ground re-replay.

Phase 3 (Years 3–4): TRL 7 → TRL 8–9

Cost: $1.5M–$3M, structured as ROSES or NASA prime contract.

- Operational deployment of TDM as the threshold-detection layer for Moon Base Phase 01 autonomous subsystems.
- Public NASA-facing Lean 4 proof browser, allowing ground operators to inspect any flight verdict on demand.
- Cross-mission adoption: Europa Clipper Whitney-class predictions, Enceladus formal prediction delivery (Khawaja et al. 2025 framework).

5 Risk matrix

Risk	Likelihood	Impact	Mitigation
Lean 4 binary too large for embedded flash	Low	High	AOT cache prunes Mathlib4 dependencies to actually-invoked theorems (< 12 MB demonstrated on benchtop)
Runtime latency exceeds 10 ms budget	Med	Med	Certificate evaluation is one Lean kernel step, not full elaboration; bench measurement < 5 ms on 200 MHz RAD750
SEU corruption of certificate buffer	Low	High	Certificates carry CRC; corrupted certificates are re-emitted, not trusted
Mathlib API churn between Lean releases	Med	Low	TDM pins to a frozen Mathlib4 release tag; upgrade is offline
CLPS provider availability for flight slot	Med	High	Phase 2 integration uses NASA-internal CLPS simulator before securing a flight slot; flight is decoupled from kernel maturity

6 Interfaces

6.1 cFS Software Bus subscription

TDM subscribes to the following cFS message IDs:

- `ENV_TEMP_TLM_MID` — thermal telemetry
- `ENV_PRES_TLM_MID` — pressure telemetry
- `ENV_RAD_TLM_MID` — radiation dosimeter telemetry
- `IMU_TLM_MID` — inertial / seismic telemetry

The cFS Software Bus delivers messages as CCSDS packets; TDM's input adapter decodes the CCSDS header and unwraps the sensor payload.

6.2 Publish targets

TDM publishes:

- `TDM_VERDICT_TLM_MID` — the current verdict (green / amber / red) and certificate hash.
- `TDM_CERT_DUMP_MID` — on-demand certificate dump, for ground-side audit.

6.3 Ground-side replay

The Lean 4 certificate is a 256-byte hash plus a serialised proof term. A ground operator runs `lake exe tdm_verify <cert_file>` to re-check the verdict. This replay is independent of TDM's flight binary — it uses the same Mathlib4 release tag but a fresh Lean toolchain on the ground.

7 Public/private partnership model

The proposed split (RFI §9):

G6 LLC contributes	NASA contributes	Joint deliverable
TDM kernel (MIT-licensed, open source)	cFS architecture guidance + interface specifications	TDM as a cFS application in the cFS catalog
21 Zenodo deposits (open access)	CLPS public telemetry (VIPER, etc.)	Lean 4 training materials
PI time (100% effort)	Technical review from JSC / JPL scientists	Peer-reviewed publication

IP terms: TDM software is MIT-licensed; mathematical framework is CC BY-NC-ND 4.0. NASA receives full use rights to all deliverables.

8 Why this matters for Moon Base Phase 01

Threshold-detection latency budgets at the Moon's South Pole are constrained by the 354-hour lunar night and Earth–Moon round-trip light time. A detection layer that operates on local geometry alone, with a formally verified safety margin, removes the Earth-call dependency for routine threshold events. This shifts the operational concept from "request authorisation from Earth" to "execute locally and log the proof certificate" for the entire class of routine threshold detections.

For Moon Base Phase 01, the projected benefit is:

- Increased autonomy for habitat thermal management during lunar night.
- Real-time qualification of ISRU regolith feedstock for sintering.
- Continuous safety margin reporting for radiation-hardened electronics in extended uncrewed mode.

9 What G6 LLC has done already

- 21 Zenodo deposits in the Principia Orthogona series (series root DOI `10.5281/zenodo.19117399`).
- AXLE Lean 4 repository (`github.com/TOTOGT/AXLE`) with > 200 theorems verified, zero `sorry` in the core results as of June 25, 2026 (see `OPEN_QUESTIONS.md` and `proofs/sorry_closures.pdf`).
- G6 Crystal (`github.com/TOTOGT/geometry`) — 35 theorems mapping NASA functional gap codes (FN-H-101L through FN-A-401L) to verified Lean proof obligations.
- Enceladus proposal (Khawaja et al. 2025 reinterpretation as Gronwall basin selection) — complete and available on request.

10 Asks of NASA

- Place G6 LLC on the LEIA Industry Day notification list (already requested in the June 2 capability statement).
- Identify a NASA technical reviewer (JSC or JPL) for the TDM kernel and Lean 4 certificate format.

- Authorise a Space Act Agreement or purchase order for the quick-start option (\$25K–\$75K, 3–6 months), under which G6 LLC delivers a written threshold-event modelling consultation plus a TOGT–ADD gap-code cross-walk appendix.